

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. Examination (CE)

May 2013

CE402 Advanced Operating Systems

Date: 10.05.2013, Friday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) How IPC will work using files? Is this safe method to perform IPC? Justify your answer with proper example. [04]
- (b) Differentiate between process and thread with proper example. [03]
- Q - 2 (a) What do you mean by distributed system? Explain basic types of distributed system. [06]
- (b) "Processes can not directly access the kernel" True/False? Justify your answer. [04]
- (c) Which are the basic types of process? Draw and explain process address space component. [04]

OR

- Q - 2 (a) Differentiate between token based and non-token based mutual exclusion. Explain any one non-token based mutual exclusion algorithm. [06]
- (b) Do as directed: [04]
- (i) A process which has just terminated but has yet to relinquish its resources is called. Give a reason to support your answer.
1. A suspended process
 2. A zombie process
 3. A blocked process
 4. A terminated process
- (ii) When a process requests for I/O, how many process switches take place? explain your answer.
- (c) How to make synchronization in distributed system? [04]
- Q - 3 (a) Answer the following: (Any Three) [12]
- (1) In Distributed System, What is the need of Election Algorithm? What is the goal of Election Algorithm? Explain "A Ring Election Algorithm".
 - (2) Explain u-are and proc data structure.
 - (3) Describe the differences between symmetric and asymmetric multiprocessing.
 - (4) Suppose a process does not wish to block until its children terminate. How can it ensure that child processes are cleaned up when they terminate? Explain in detail.
- (b) In what situations is an application better off using multiple processes rather than LWPs or user threads? [02]

SECTION – II

- Q - 4 (a) Attempt the following:** [04]
- (1) Briefly explain atomicity and message ordering in group communication.
 - (2) When an input file is opened, what are the possible errors that may occur?
- (b) What is Client Stub in remote procedure call ? Explain with an example.** [03]
- Q - 5 Answer the following (Any Three)** [14]
- (1) What is the sequence of events during remote procedure call?
 - (2) Differentiate between message passing and remote procedure call.
 - (3) Explain upload/download model in the File Service Interface.
 - (4) What is shared memory? Explain bus-based multiprocessor.
- Q - 6 (a) Answer the following (Any Three)** [12]
- (1) Why remote procedure call (RPC) doesn't fit in OSI model?
 - (2) What are the various way of doing caching in main memory? Explain any one strategy in detail.
 - (3) List out various consistency model used in DSM (Distributed Shared Memory). Explain any one in detail.
 - (4) What are the main differences between a LAN and a WAN? Why are medium access control protocol needed?
- (b) Explain context switch between two threads in the same process.** [02]
